SRS report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Content Quality	2					
Clarity	1					
Spelling & Grammar	1					
Organization and Formatting	1					
					Total obtained mark	
Comments						

Semester: Spring 2026

Student ID: 0242310005341201 Student Name: SHAH SHAZID ASKARY

Student ID: 0242310005341163 Student Name: SUDIPTO BISWAS

Student ID: 0242310005341046 Student Name: Masud Hassan Sakib

Batch: 40 Section: E1

Course Code: SE331 Course Name: Software Engineering Design Capstone Project

Course Teacher Name: Rahat Uddin Azad

Designation: Lecturer

Department: Department of Software Engineering

Submission Date: 20 / 02 / 2026

Software Requirements Specification (SRS) for TakaTracker

1. Introduction

1.1 Problem Statement

Small merchants in Bangladesh encounter significant challenges in managing their daily business transactions, tracking customer and supplier dues, and maintaining accurate financial records. This is primarily due to their reliance on traditional paper-based bookkeeping methods. Key issues identified include the potential for paper damage or loss, frequent calculation errors, limited internet access in rural areas, complex interfaces of existing digital solutions, and language barriers. These factors collectively hinder efficient business operations and accurate financial oversight.

1.2 Purpose

The primary purpose of the TakaTracker project is to develop a user-friendly and offline-capable mobile application. This application aims to digitize traditional bookkeeping practices for Bangladeshi small merchants, enabling them to efficiently track transactions, manage customer dues, and gain valuable insights into their business performance. The goal is to provide a robust and accessible tool that simplifies financial management for micro and small businesses.

1.3 Scope

In-Scope Features:

The TakaTracker application will include the following functionalities:

- **User Authentication**: Phone number-based registration and login with OTP verification and JWT for secure sessions.
- **Dashboard**: A comprehensive daily summary displaying sales, receivables, and payables.
- **Transaction Management**: Features for creating, editing, and deleting sales, purchases, and expense records.
- **Contact Management**: Tracking of customer and supplier dues.
- **Inventory Management**: A product catalog with stock level tracking and alerts.
- **Payment Recording**: Functionality for recording incoming and outgoing payments, including QR code generation for payments.
- **Offline Operation**: Ability to perform data entry and access records without an active internet connection.
- **Data Synchronization**: Automatic synchronization of data with the server once an internet connection is re-established.

- **Multilingual Support**: Full support for both Bengali and English languages, allowing users to switch between them.

Out-of-Scope Features:

The following features are explicitly excluded from the initial release of the TakaTracker application:

- **Advanced Accounting**: Complex functionalities such as GST/VAT calculations.
- **Multi-branch Management**: Support for managing multiple business branches.
- **Payroll Management**: Features related to employee payroll.
- **E-commerce Integration**: Direct integration with e-commerce platforms.
- **Bank Account Integration**: Actual integration with bank accounts; only stubs will be used for payment gateways.
- **Barcode Scanning**: Functionality for scanning barcodes for inventory management.
- **iOS Deployment**: The initial phase will focus exclusively on Android deployment.

2. Design and Implementation Constraints

2.1 Programming Languages & Tech Stack

Category	Technology	Justification
Frontend	React Native (Expo)	Cross-platform development for Android, enabling a single codebase and faster development cycles.
	TypeScript	Enhances code quality and maintainability through static typing.
Backend	Python 3.11+	Chosen for its versatility, extensive libraries, and suitability for rapid API development.
	Django 5.0	A high-level Python web framework that encourages rapid development and clean, pragmatic design.
	Django REST Framework	Provides powerful and flexible tools for building Web APIs.

2.2 Database & Servers

Component	Technology	Justification
Primary Database	PostgreSQL 15	Robust, open-source relational database known for data integrity, advanced features, and scalability.
Local Storage	SQLite (via AsyncStorage)	Embedded relational database for efficient offline data persistence on mobile devices.

Component	Technology	Justification
Cache Layer	Redis	In-memory data structure store, used for caching session data and implementing rate limiting to enhance performance.
API Protocol	RESTful HTTP/HTTPS	Standard, stateless communication protocol for flexible and scalable API interactions, ensuring secure data transmission.

2.3 Development & Deployment Constraints

- **Target Operating System**: Android API 21+.
- **Offline-First Architecture**: The application must function effectively without an internet connection, with data synchronizing automatically when connectivity is restored.
- **Security**: All data transmission will be secured using HTTPS. Sensitive user data, such as passwords, will be hashed. OTP will be used for user verification, with a defined expiration time.
- **Timezone**: The system will operate in the Asia/Dhaka (UTC+6) timezone.
- **Currency**: The default currency for all transactions will be BDT (Bangladeshi Taka).

3. User Classes and Characteristics

3.1 User Roles and Descriptions

User Class	Description	Technical Skill	Primary Language	Permissions/Key Actions
Small Merchants	Business owners (shopkeepers, traders, freelancers) managing daily transactions.	Low to moderate. Comfortable with smartphone apps.	Primarily Bengali.	Full CRUD operations on transactions, contacts, inventory. Track dues, view reports.
Suppliers	Entities that provide goods or services to small merchants.	N/A	N/A	Track credit extended to merchants.
Shop Staff (optional)	Employees who record transactions on behalf of the merchant owner.	Basic.	N/A	Record transactions as directed by the merchant.
System Administrators	Personnel responsible for maintaining the backend system.	High.	English.	Backend maintenance, data management, support.

4. Functional Requirements (FR)

ID	Requirement Name	Description	Users	Priority
FR-1	User Authentication	The system shall allow users to register and log in using their phone number and OTP verification.	Small Merchants	High
FR-2	Dashboard Display	The system shall display a daily summary of total sales, total profit, and total buying price along with a GitHub-style activity graph.	Small Merchants	High
FR-3	Transaction Management	The system shall allow users to create, edit, and delete sales, purchase, and expense records.	Small Merchants	High
FR-4	Contact Management	The system shall enable tracking of customer and supplier dues.	Small Merchants	High
FR-5	Inventory Management	The system shall provide a product catalog and alert users about low stock levels.	Small Merchants	Medium
FR-6	Payment Recording	The system shall allow users to record incoming and outgoing payments and generate QR codes for payment collection.	Small Merchants, Suppliers	High
FR-7	Offline Operation	The system shall allow all core functionalities to be used without an active internet connection.	Small Merchants	High
FR-8	Data Synchronization	The system shall automatically synchronize local data with the server when an internet connection is available.	Small Merchants, Admin	High
FR-9	Multilingual Support	The system shall allow users to switch between Bengali and English languages.	Small Merchants	Medium
FR-10	Notification	The system shall send notifications when product inventory reaches a low-stock threshold or becomes out of stock, notifying the merchant and supplier for restocking.	Small Merchants, Suppliers	Medium

5. Non-Functional Requirements

5.1 Performance

- The dashboard and key data displays shall update within 2 seconds of data changes or retrieval.
- The synchronization engine shall efficiently resolve data conflicts during offline-to-online transitions to ensure data consistency.

5.2 Security

- All sensitive data, including user credentials and financial records, shall be encrypted using AES-256 standards (or equivalent strong encryption) both in transit and at rest.
- User authentication will be secured using JSON Web Tokens (JWT) with appropriate expiration and refresh mechanisms.
- All communication between the mobile application and the backend server shall be conducted over HTTPS.
- Passwords will be stored as hashed values, not in plain text.

5.3 Usability

- The user interface shall be designed for simplicity and ease of use, catering to users with low to moderate technical proficiency.
- The application shall feature high-contrast elements and legible fonts to ensure accessibility for all users.
- The primary language will be Bengali, with English as an alternative, to ensure comfort and familiarity for the target user base.

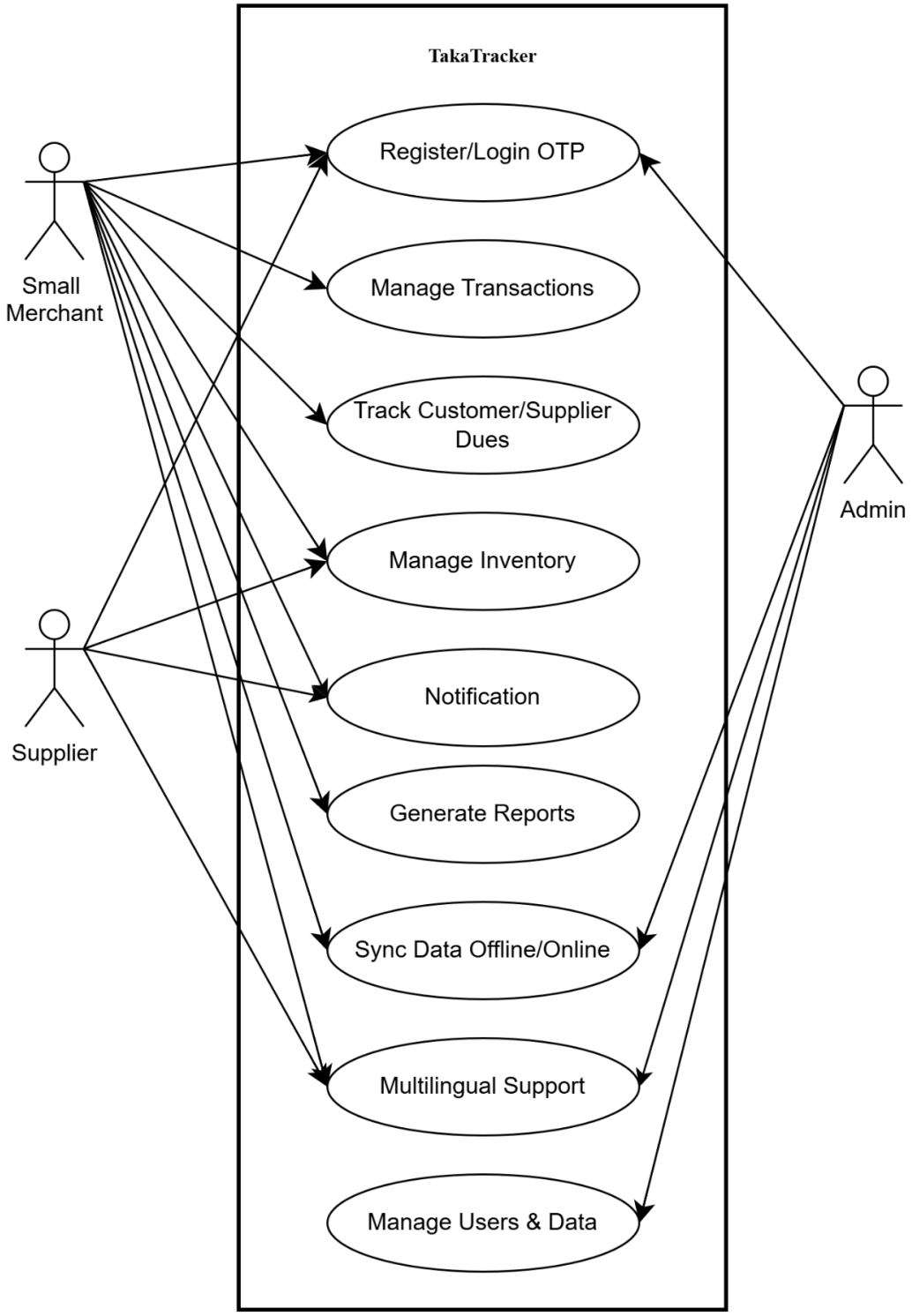
5.4 Reliability

- The backend system shall maintain an uptime of at least 99%.
- The local data persistence mechanism (SQLite) shall ensure that no data is lost during offline operations or unexpected application closures.

6. System Design Diagrams

This section provides visual representations of the system logic.

6.1 Use Case Diagram



6.2 Use Case Descriptions (Main Success Flow – High Priority Features)

FR-1: User Authentication

Primary Actor: Small Merchant

Main Success Flow:

1. The user opens the application.
2. The system prompts the user to enter a phone number.
3. The user enters their phone number.
4. The system sends an OTP to the provided phone number.
5. The user enters the received OTP.
6. The system verifies the OTP.
7. The system creates a new account (if first-time user) or logs the user in.
8. The dashboard is displayed.

FR-2: Dashboard Display

Primary Actor: Small Merchant

Main Success Flow:

1. The user logs into the system.
2. The system retrieves sales, profits, and buying price data.
3. The system calculates the daily summary.
4. The dashboard displays:
    - Total Sales
    - Total Profit
    - Total buying price
    - Github style graph
5. The user views the summarized financial information.

FR-3: Transaction Management

Primary Actor: Small Merchant

Main Success Flow:

1. The user navigates to the Transactions section.
2. The user selects “Add Transaction.”
3. The system displays transaction form (sales/purchase/expense).
4. The user enters transaction details.
5. The user saves the transaction.

6. The system validates the data.
7. The system stores the transaction in local storage.
8. The dashboard is updated automatically.

FR-4: Contact Management

Primary Actor: Small Merchant

Main Success Flow:

1. The user opens the Contacts section.
2. The user selects “Add Customer” or “Add Supplier.”
3. The user enters contact details.
4. The system saves the contact.
5. The user records transactions linked to the contact.
6. The system automatically updates receivables/payables.
7. The user views updated dues status.

FR-5: Inventory Management

Primary Actor: Small Merchant

Main Success Flow

1. The user navigates to the Inventory section.
2. The user selects “Add Product.”
3. The system displays the product form.
4. The user enters:
    - Product Name
    - Category
    - Price
    - Stock Quantity
5. The user saves the product.
6. The system validates the entered information.
7. The system stores the product in local storage.
8. The product appears in the product catalog list.

FR-6: Payment Recording

Primary Actor: Small Merchant

Main Success Flow:

1. The user navigates to the Payments section.
2. The user selects “Record Payment.”
3. The user selects related customers/suppliers.
4. The user enters payment amount and method.
5. The system updates outstanding dues.
6. The system saves payment records.
7. The system optionally generates a QR code for payment collection.
8. The dashboard updates automatically.

FR-7: Offline Operation

Primary Actor: Small Merchant

Main Success Flow:

1. The user opens the application without internet connection.
2. The system detects no active internet connection.
3. The system enables offline mode.
4. The user performs core actions (transactions, payments, contact management).
5. The system stores all data locally.
6. The system marks records as “Pending Sync.”

FR-8: Data Synchronization

Primary Actor: Small Merchant

Supporting Actor: Admin

Main Success Flow:

1. The system detects internet connectivity.
2. The system automatically triggers synchronization.
3. Local pending data is prepared for upload.
4. The system sends data to the server.
5. The server validates and stores data.
6. The server sends a confirmation response.
7. The system marks records as synchronized.
8. The user receives sync success notification.

FR-9: Multilingual Support

Primary Actor: Small Merchant

Main Success Flow:

1. The user navigates to Settings.
2. The user selects “Language Option.”
3. The system displays available languages (Bengali / English).
4. The user selects their preferred language.
5. The system updates the interface language immediately.
6. All menus, labels, and notifications are displayed in the selected language.
7. The system saves language preference for future sessions.

FR-10: Inventory Alert Notification

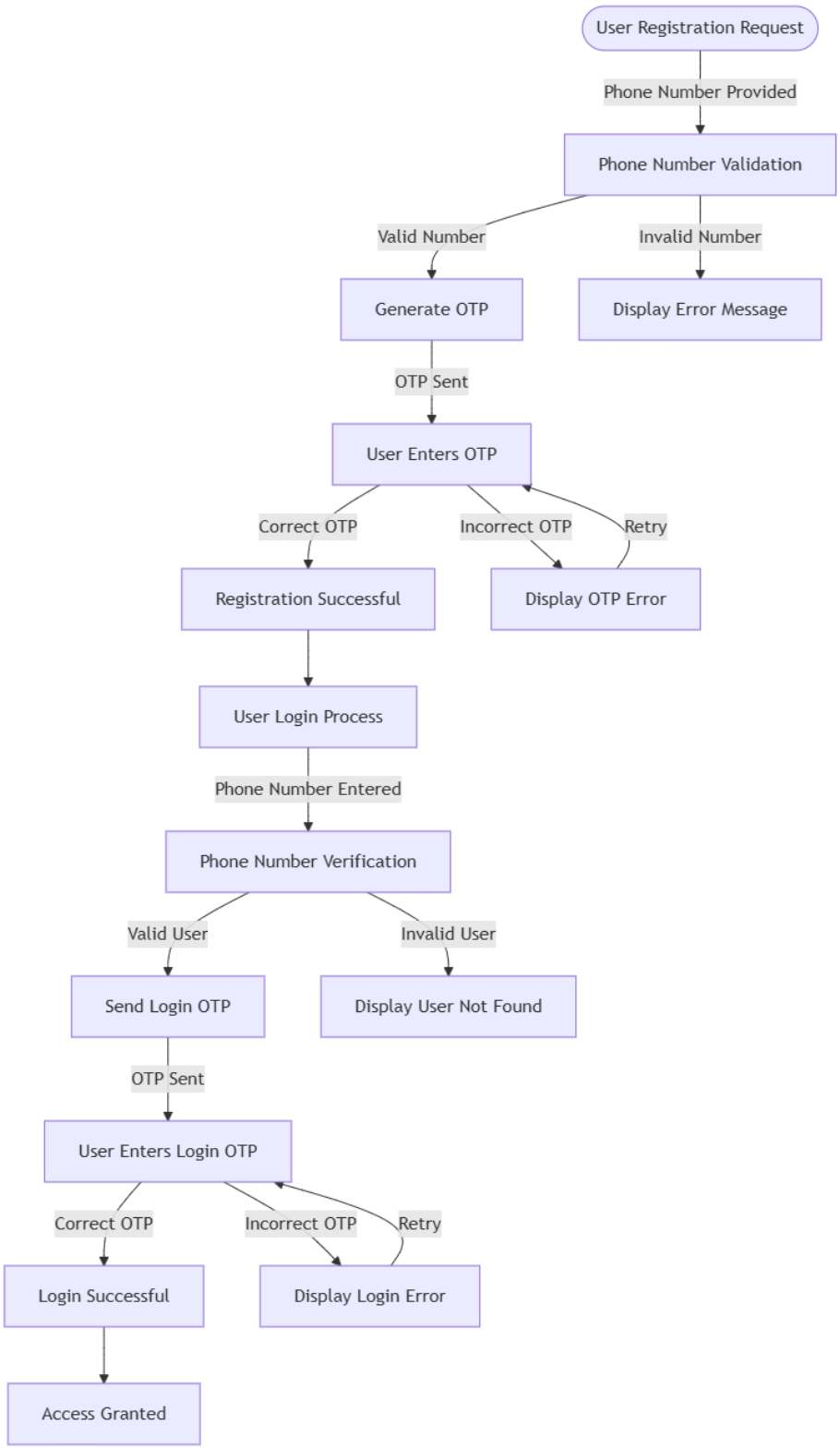
Primary Actor: Small Merchant
Supporting Actors: Customer, Supplier

Main Success Flow:

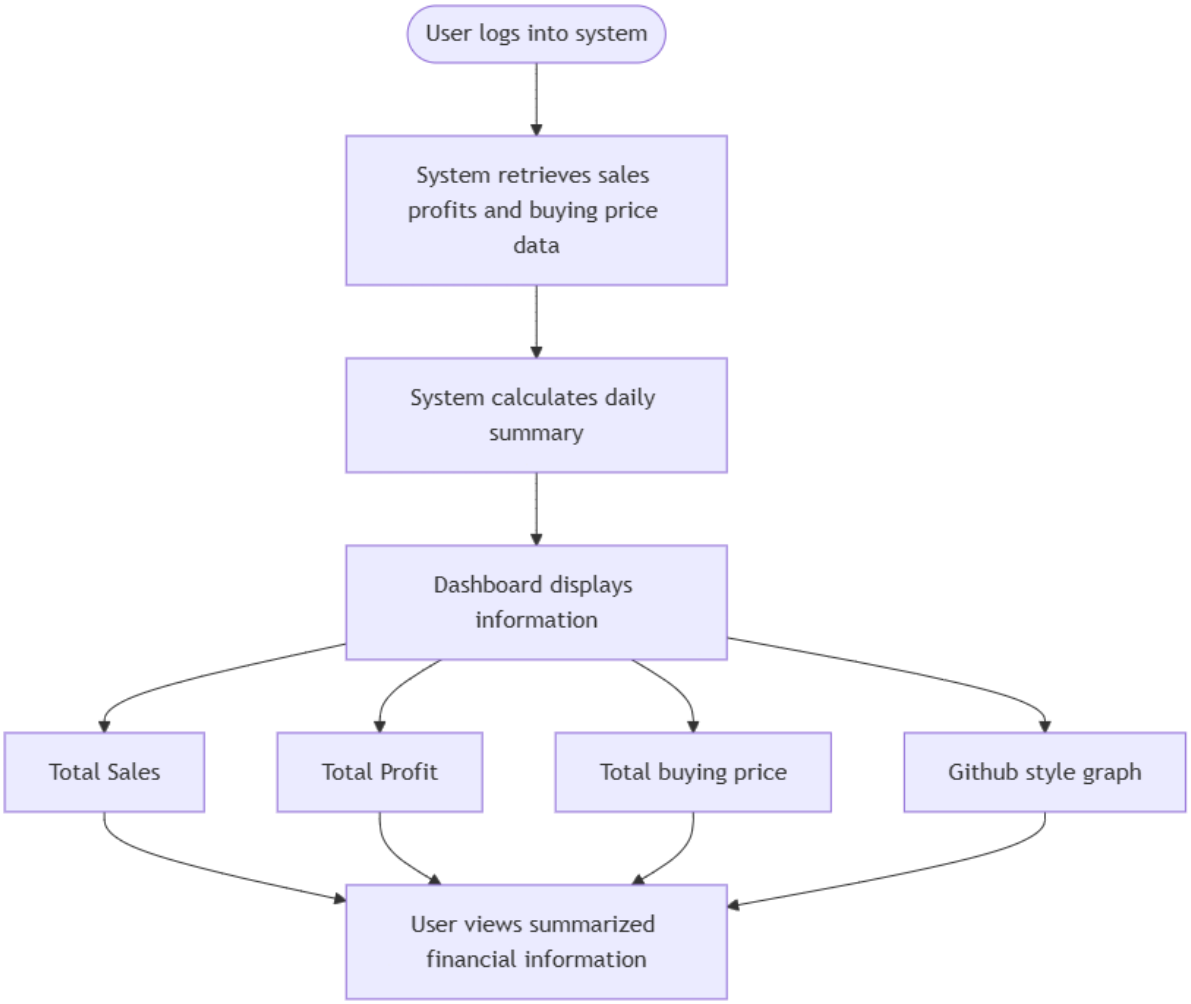
1. The system continuously monitors product stock levels in the inventory.
2. When the stock quantity of a product reaches the predefined minimum threshold or becomes zero, the system detects the low inventory status.
3. The system automatically generates a Low Stock / Out of Stock notification.
4. The Small Merchant receives a notification on the dashboard indicating that the product inventory is finished or running low.
5. The Customer receives a notification informing them that the product is currently out of stock or temporarily unavailable.
6. The Supplier receives a notification requesting restocking or supply of the product.
7. The system records the notification event in the notification history.
8. Once the merchant updates the inventory with new stock, the system automatically removes the out-of-stock alert and updates product availability status.

6.3 Activity Diagram

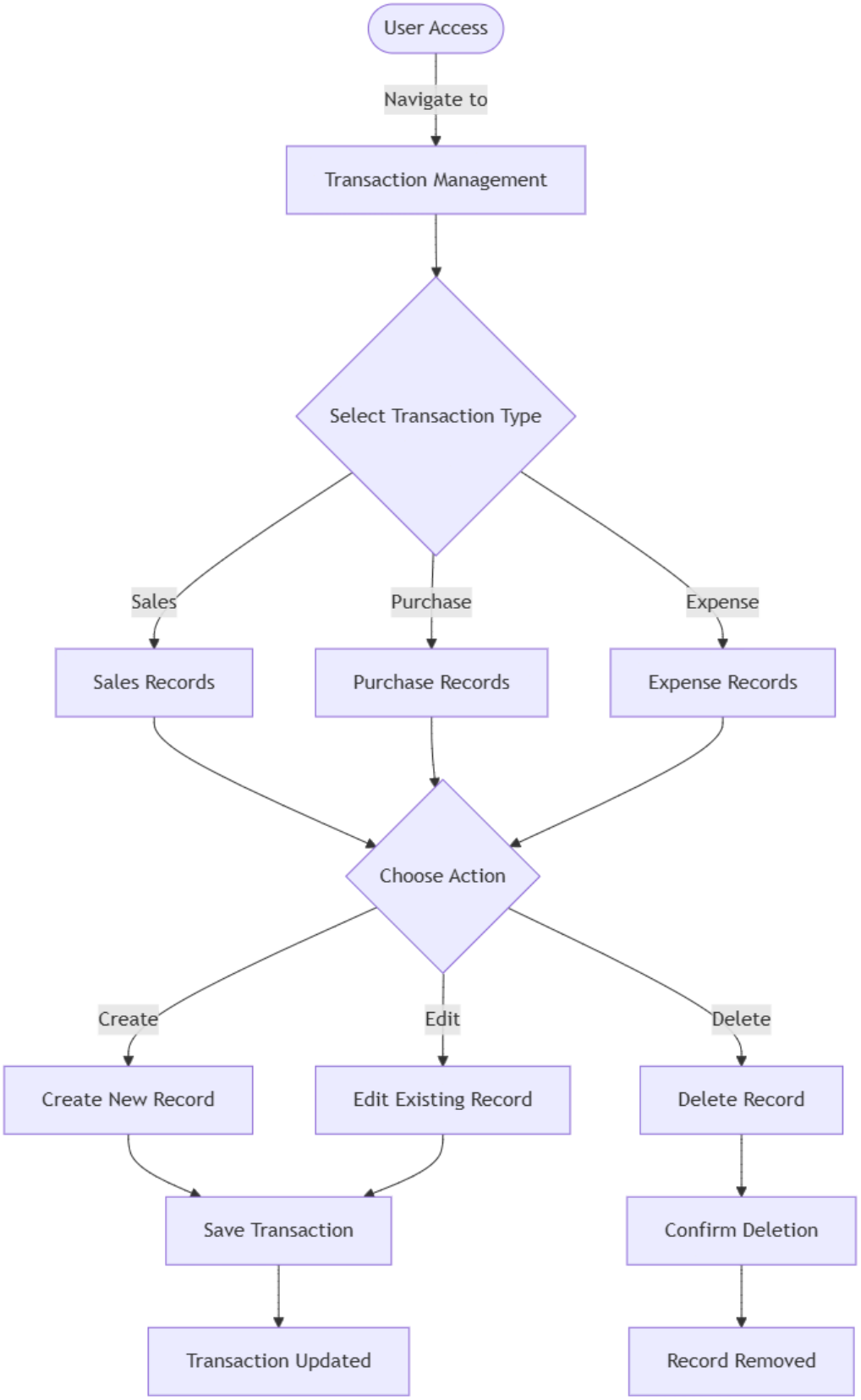
FR - 1 User Authentication



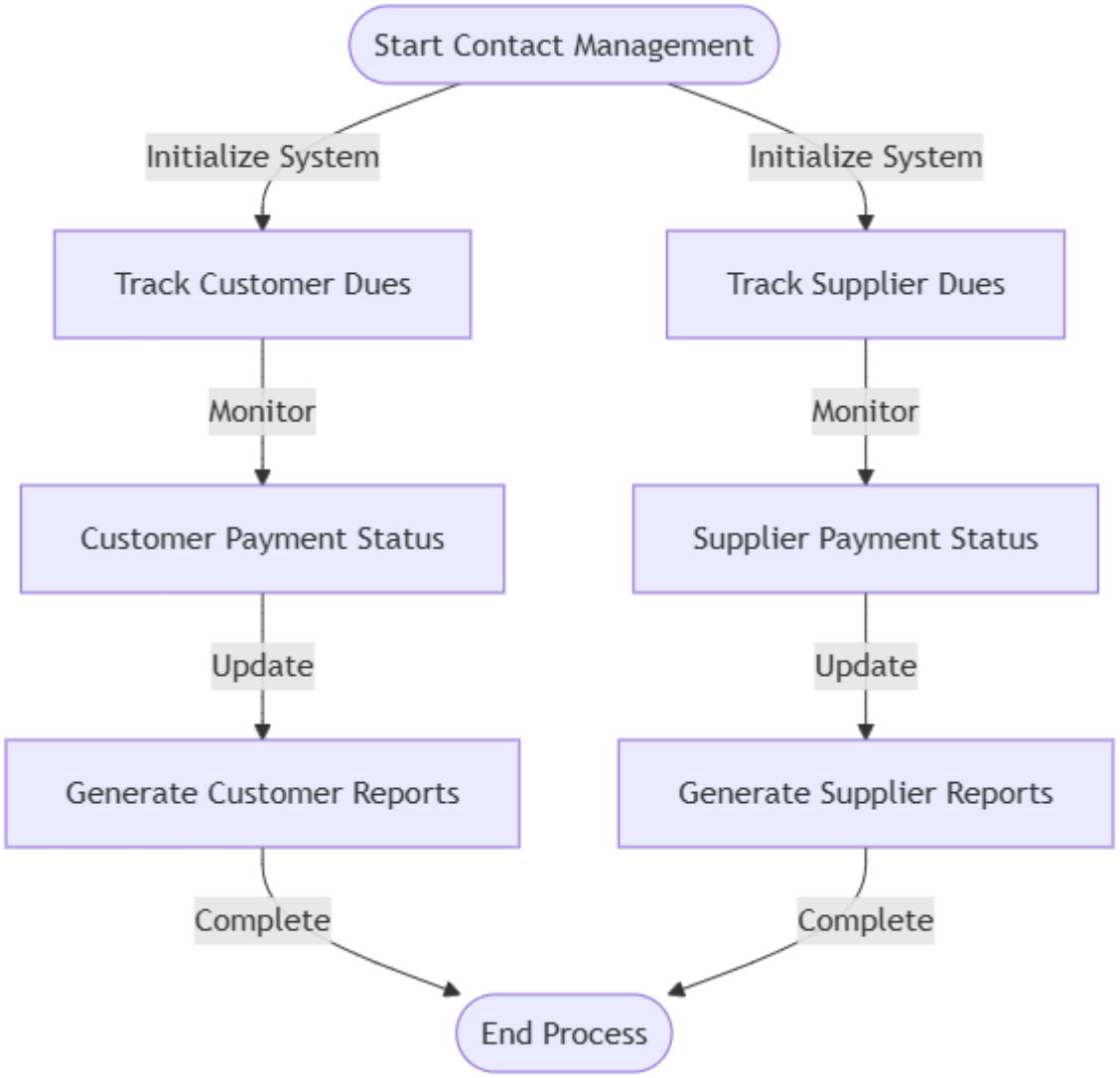
FR - 2 Dashboard Display



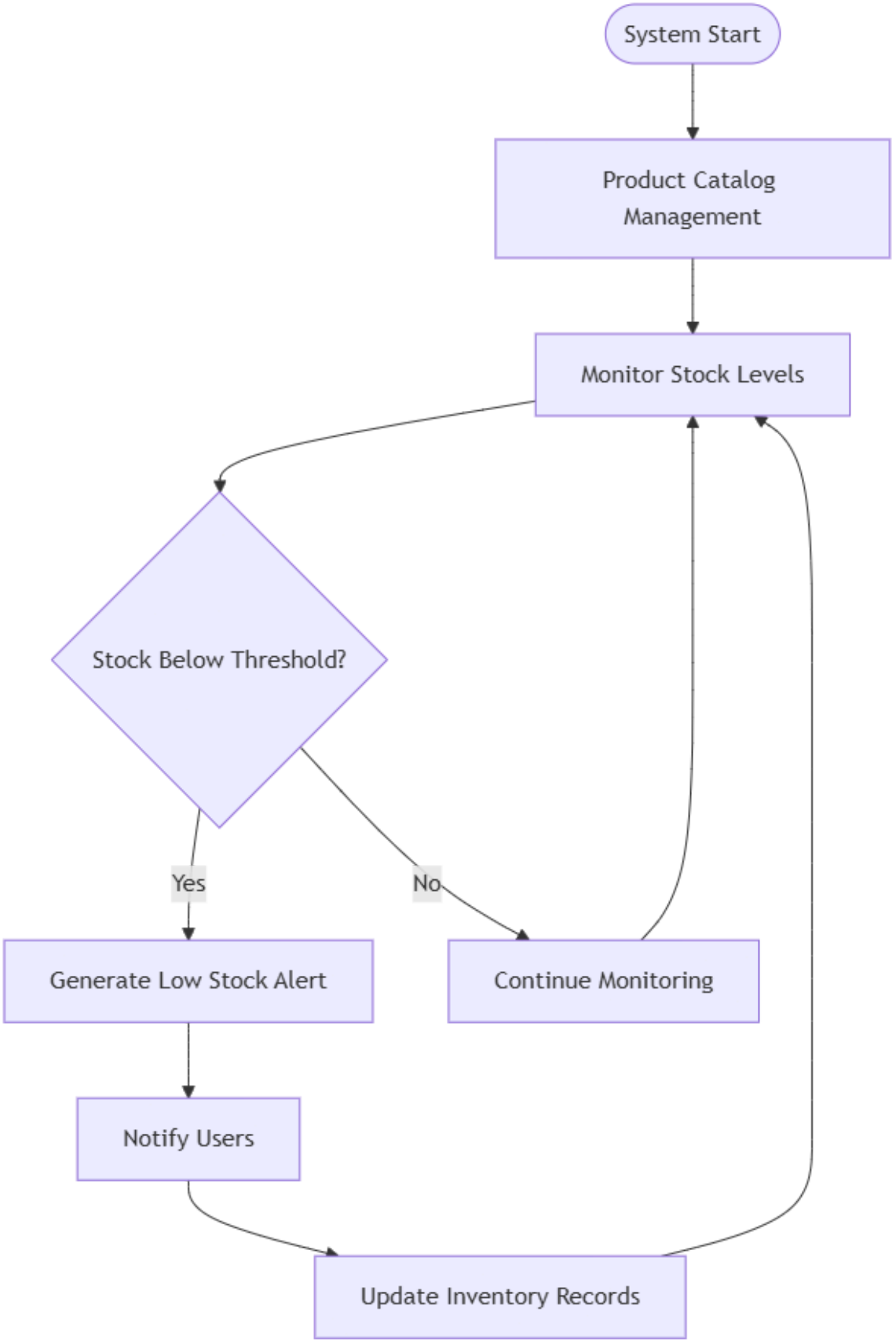
FR - 3 Transaction Management



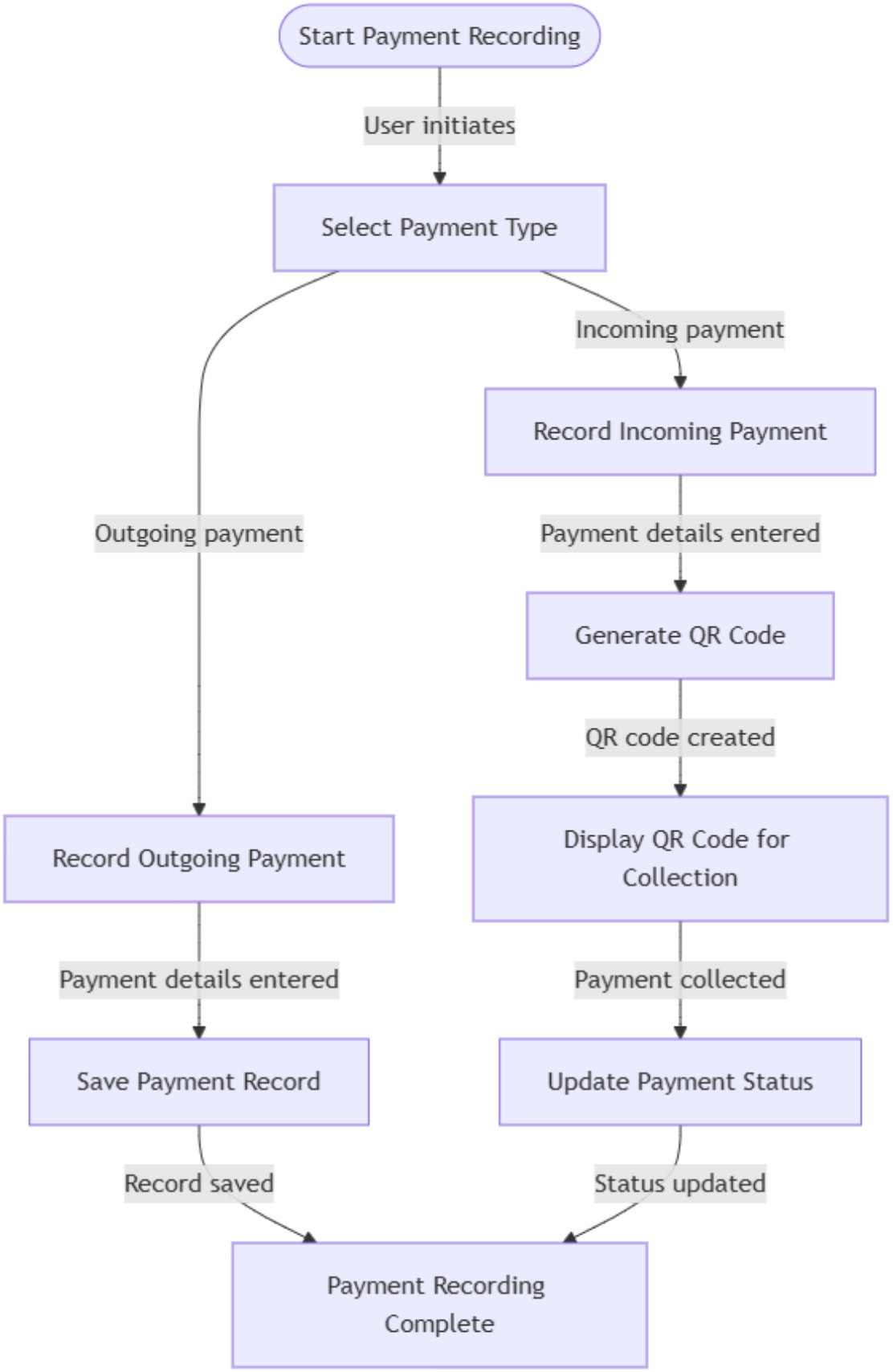
FR - 4 Contact Management



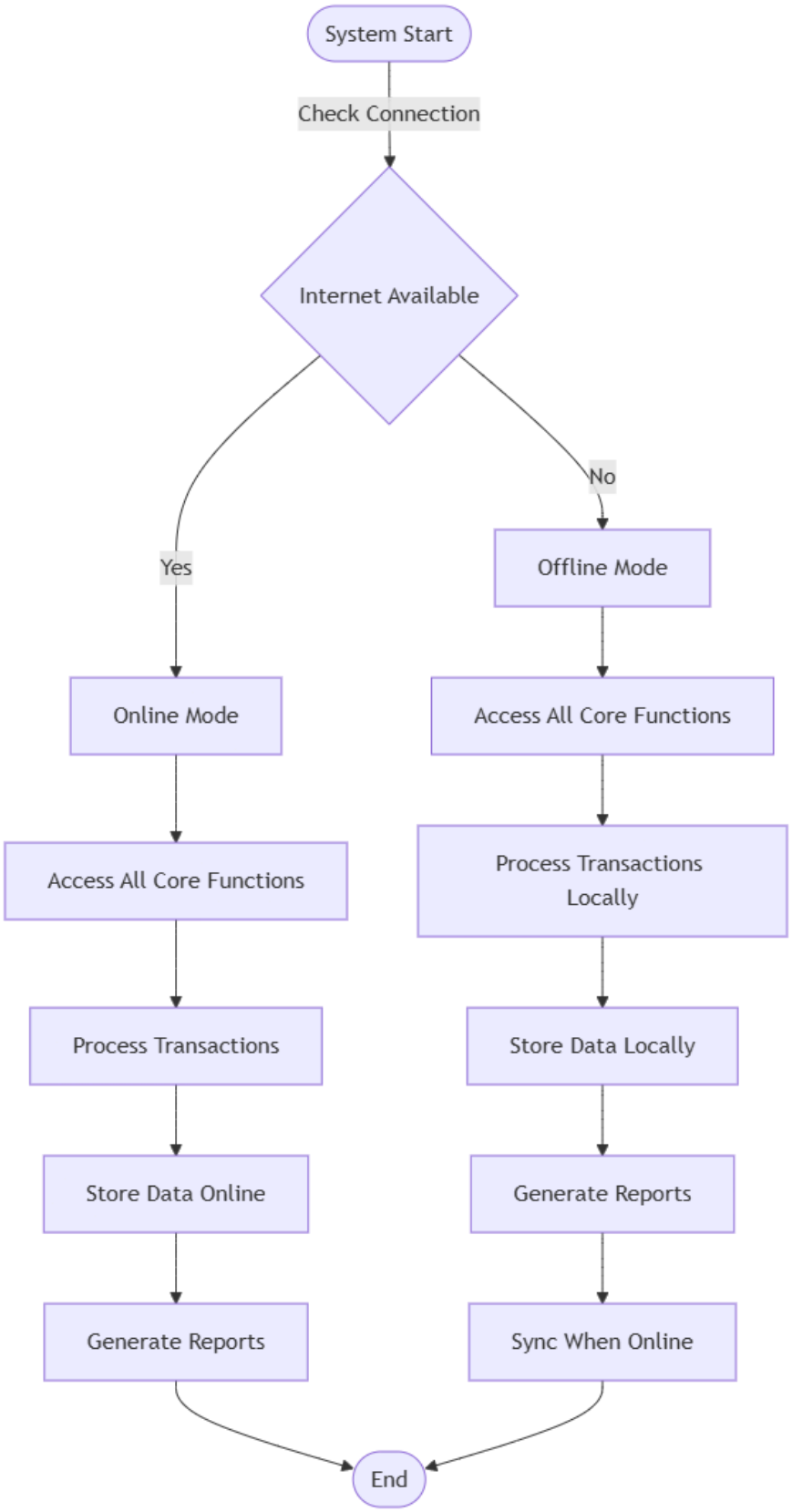
FR - 5 Inventory Management



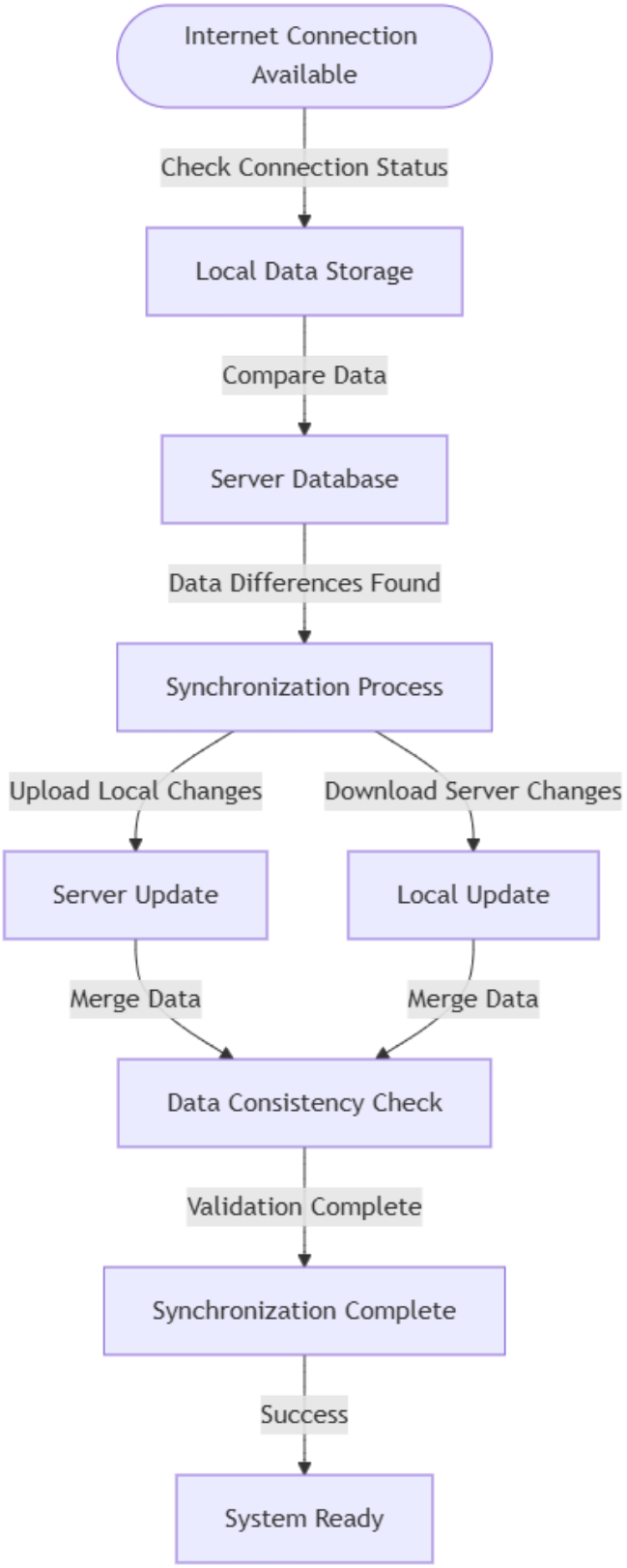
FR - 6 Payment Recording



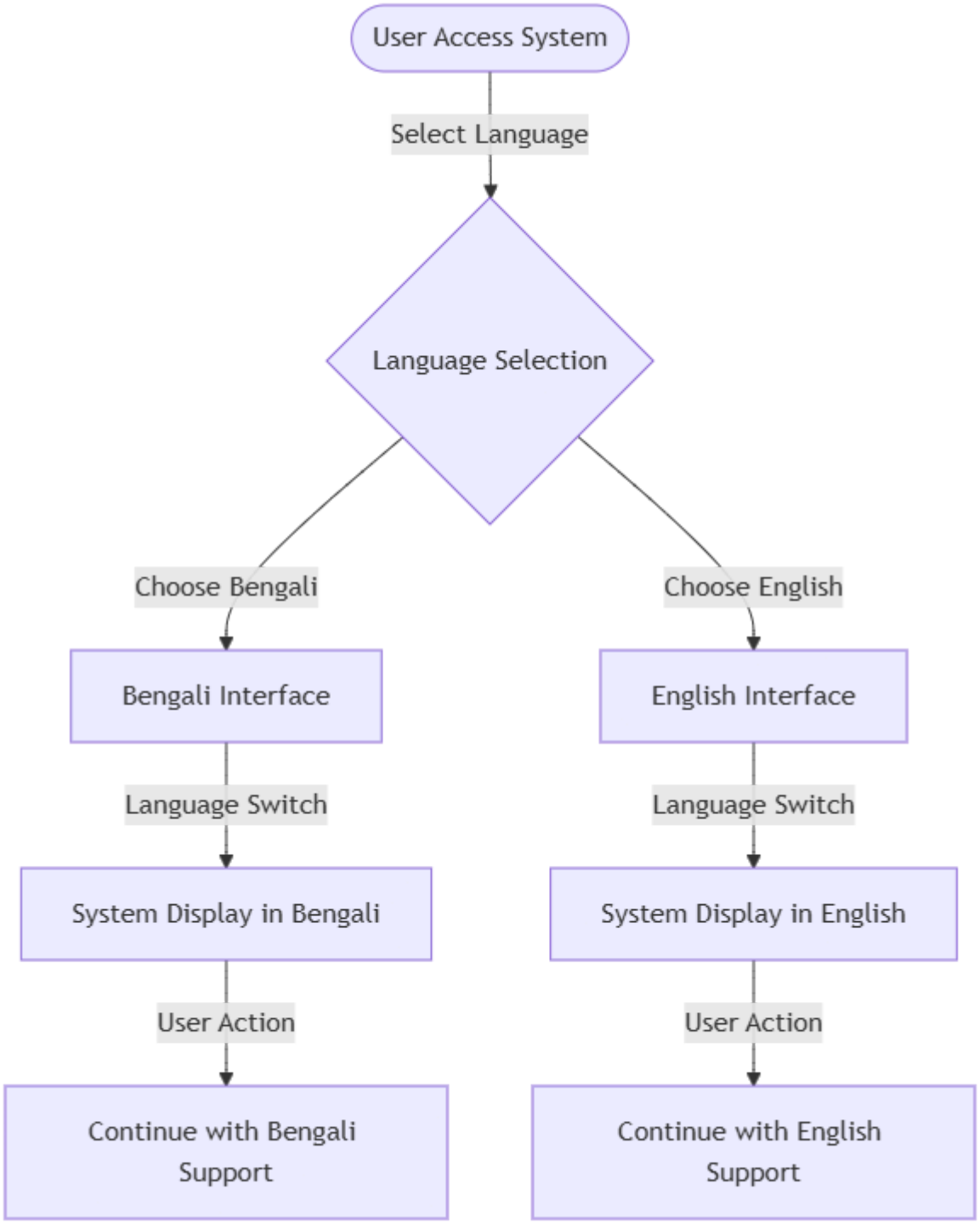
FR - 7 Offline Operation



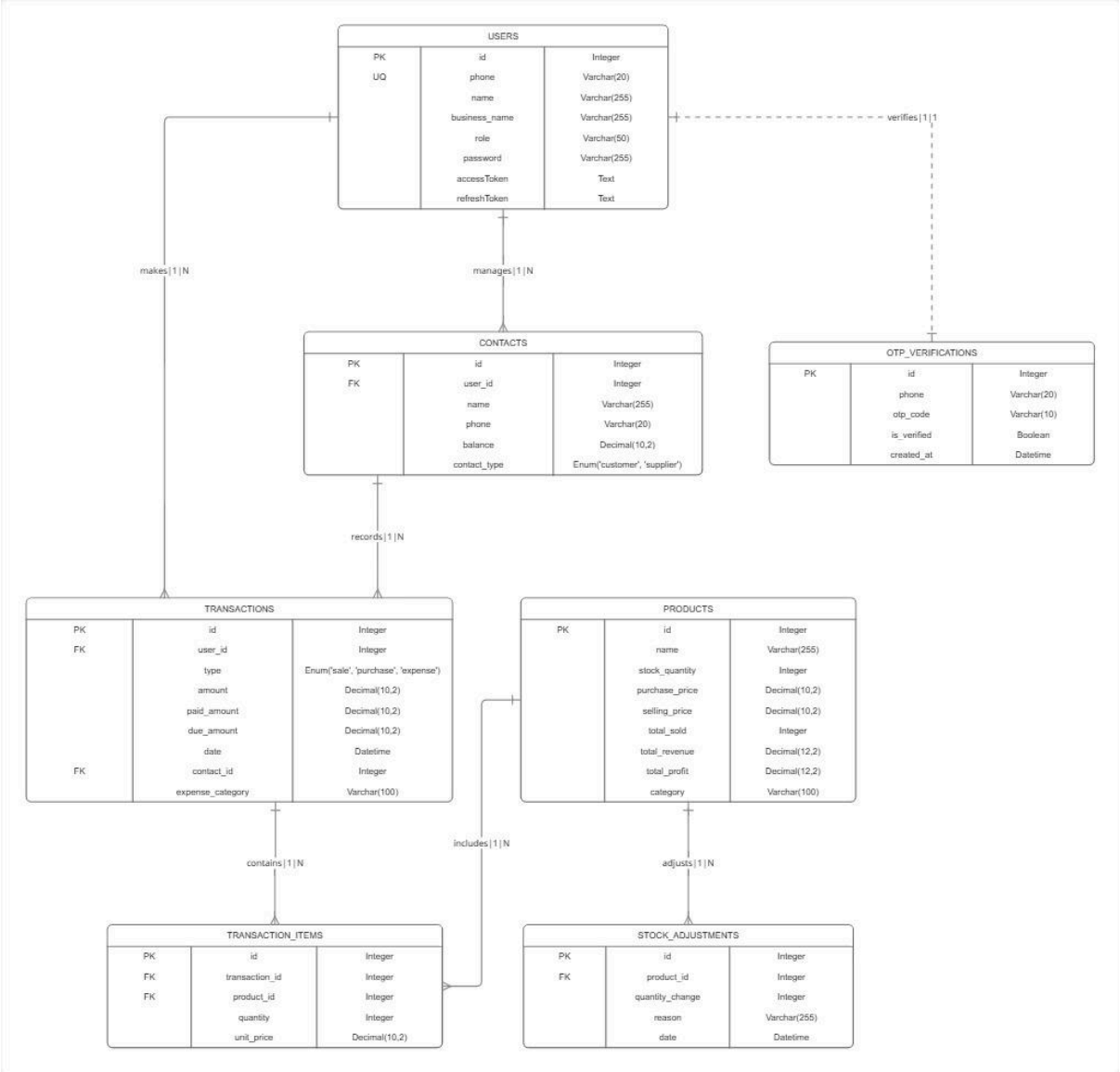
FR - 8 Data Synchronization



FR - 9 Multilingual Support



6.4 Entity Relationship Diagram (ERD)



7. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Sudipto Biswas	0242310005341163	Front-end UI/UX & State Management.
Shah Shazid Askary	0242310005341201	Backend API development & Database Schema design.
Masud Hassan Sakib	0242310005341046	Sync Engine & Offline Logic development.